DRUMPANEL · FEASIBILITY

DrumPanel feasibility assessment

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DOCUMENT CONTROL

Document ID	DMP-FEA-001	Version	0.1
Status	Draft	Date	2026-10-24
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/drumpanel		
Source	docs/08-feasibility.md @ de21d74		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First feasibility assessment (kit 1.8 upgrade of a TRL 3 repo), closed for TRL 3 from DMP-STM-001, DMP-CAL-001, the D3 model and the build plan

DrumPanel feasibility assessment

Feasible on paper and closed for TRL 3: every station is sawn, drilled and stick-welded from square tube and plate, with three rolls turned on a lathe and a handful of bought precision parts, and every crank and lever force is within 150 N. Two things are not settled by paper: throughput (about 2.20 drums an hour against the 2.5 of PR-007) and the screening coefficients behind the hardest forces (the chime notch, the disc traction), which the TRL 4 trial measures. Cost is USD 3,566 against the USD 3,000 value-engineering target. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	The purge, vapour check, end cutter, notching punch, rotary shear head with ring head and slip roll each meet their force and fit checks on paper; the D3 model of 818 parts has no interference; the shear head clears the drum in slit and ring modes	High	Two ring cuts need 136 N against 150 N; the chime notch force (24.3 kN) and the disc traction (1.17) rest on screening values (DMP-STM-001 S6, S7, S13)	DMP-CAL-001 sections 3 to 9; DMP-STM-001; `cad/fidelity.json`
Materials	S275 tube and plate, C45 rolls, D2 discs and punches hardened by a tool shop, bronze bushes, hardwood chocks, plywood tables	High	Drum steel varies by batch (yield 200 to 300 MPa, wall 0.8 to 1.2 mm), which is why the bending roll is set by a trial pass for each batch and the first panel of every drum is checked (STEM fix F1)	DMP-CAL-001 Table 1 and section 5.3
Manufacturing and assembly	Welding, drilling, tapping and turning; the rolls need a lathe with 1.1 m between centres; 18 assembly steps; the D3 detailing changed only fixings and small sizes (DMP-DDR-006)	Medium	A long lathe may be in the next town; the punch dies and discs come from a tool shop	DMP-BLD-001; DMP-DWG-001 and DMP-DWG-101 to 123
Power	Human powered: about 20 to 136 N at the handles; no electricity, fuel or compressed air	High	Repetitive cranking over a day (DMP-RSK-001 H7)	DMP-CAL-001 sections 3 to 9; PR-015
Controls	None; the bending roll is set by dials, the shear head by an index pin, the drum by a screw brake	High	A wrong dial reading leaves panels out of flat; the trial pass catches it	DMP-CAL-001 section 5.3
Supply of parts	Steel, fasteners, bearings, bushes, sprockets, chains and gears are commodity parts; the gas detector, its bump-test gas and the discs need specialist suppliers	Medium	Calibration gas and detector service may not be available locally (DMP-RSK-001 P4)	BOM lines 4, 10, 16, 21, 25; DMP-MKT-001

Ratings: High, Medium, Low.